

Body parts used

- **Swimming bouts:** uses whole-fish signals derived from **all body parts** (Summed_movement, Bp_velocity_mean).
 - **Turning metrics:** orientation from **bodypart2** → **bodypart16**.
 - **Path metrics:** **bodypart16** only.
 - **Energy metrics:** per-part velocity → **all body parts**.
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1) Swimming bouts

These columns mark active vs. inactive swimming, label bouts, and give bout-level stats.

- **Active_swimming**
1 if the frame's Summed_movement is above a threshold ($\text{mean} \times 1.5$), else 0.
Units: binary.
- **Bout_ID**
Integer label that stays constant while Active_swimming stays the same, then increments at each change.
Units: integer ID.

For each frame, the next three columns copy the stats of the **current bout** onto the frame (handy for plotting / ML):

- **Current_bout_duration**
Duration of the current bout up to now.
Units: **seconds**.
- **Current_Summed_movement_mean**
Mean Summed_movement within the current bout.
Units: **mm/frame**.
- **Current_Bp_velocity_mean_mean**
Mean of Bp_velocity_mean within the current bout.
Units: **mm in last 1 s** ($\approx \text{mm/s}$).
- **Bout_frequency_{W}**
Estimated number of **bout starts per second** over the last **W frames**.
(It rolls a count of bout starts and scales by fps / W .)

Units: Hz (bouts/s).

Example: Bout_frequency_150 $\approx$ frequency over ~5 s if 30 fps.

Body parts used: all (because Summed_movement and Bp_velocity_mean aggregate per-part signals).

2) Turning metrics

Per-frame turning speed, flags for sharp turns, and durations of turning bouts.

- **Turning_rate**

Absolute per-frame heading change (wrapped) converted to **turns per second**:

$|\Delta\text{angle_deg}| * \text{fps} / 360$.

Units: turns/s (revolutions per second).

- **Sharp_turn**

1 when the rolling sum of $|\Delta\text{angle}|$ over a short window ($\sim\text{fps}/4$) exceeds **30°**; else 0.

Units: binary.

- **Turn_bout_duration**

For frames that are inside a sharp-turn bout, this is the bout's duration (else 0).

Units: seconds.

Body parts used: bodypart2 (front/head) and **bodypart16** (tail) to form the orientation vector.

3) Path metrics

Tortuosity (how squiggly) and efficiency (straightness) of the track, computed from a single anchor.

- **Path_tortuosity_{W}**

(cumulative distance over W) / (straight-line distance start→end over W).

Range: ≥ 1 (1 = perfectly straight).

Units: dimensionless.

Example: Path_tortuosity_150.

- **Swimming_efficiency_{W}**

$1 / \text{Path_tortuosity}_{\{W\}}$.

Range: (0, 1] (1 = perfectly straight).

Units: dimensionless.

Body part used: bodypart16 (explicit in code).

4) Energy metrics

A simple kinetic-energy **proxy** built from per-part velocities.

- **{bp}_kinetic_energy_proxy**
 $0.5 * ({bp_velocity})^2$.
Units: proportional to **(mm/s)²** (not physical Joules—mass not included).
- **Total_kinetic_energy**
Sum of all {bp}_kinetic_energy_proxy across body parts (per frame).
Units: same proxy units.
- **Cumulative_energy**
Running cumulative sum of Total_kinetic_energy over time.
Units: accumulated proxy units.
- **Energy_rate_{W}**
Rolling **mean** of Total_kinetic_energy over the last **W frames**.
Units: proxy units per frame (read as “average energy proxy” over that window).
Example: Energy_rate_150.

Body parts used: all (one proxy per body part, then aggregated).